Steering a swarm that owns a proof

Simulation 1 (days and nights), setup `task003-symmetric`

LLM-free simulator; report of 16 October 2026; runs of 15 October 2026

Mirrors the Simulation 2 report of the same date: same sections, figures and equations. Same data as the 15 October version, reorga

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. An episode has 30 rounds, each a day and a night. During the day the agents act one at a time in a new random order: an agent reads q posts drawn at random from a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. At night a *controller* with a target, A0 (truth) or A2 (false), reads q_c agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (what it read already proves its target) silences it, posts the b facts from its pool that best support its target. The board is then cleared, keeping only the controller's new posts. At each dawn every remembered fact is forgotten with probability $1-\rho$ and comes back if read again. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true. The agents hold, together, a proof of A0; no single agent does, and neither pool can prove its target (Table 1).

Table 1: Facts available at the start and what each set implies, $P(\mathrm{A0}, \mathrm{A1}, \mathrm{A2} \mid \text{facts})$.

set of facts	facts	$P(\mathrm{A0}, \mathrm{A1}, \mathrm{A2})$
agents' facts, all together (24 agents, one fact each)	18 distinct	(1, 0, 0): proves A0
A0 pool (controller with target A0)	12, of which 10 new to the agents	(0.963, 0.037, 0)
A2 pool (controller with target A2)	12, of which 10 new to the agents	(0.114, 0, 0.886)
agents' facts + either pool	28 distinct	(1, 0, 0)

The runs. Posts read by an agent $q \in \{3, 6, 12\}$; posts read by the controller $q_c \in \{3, 6, 12, 24\}$; facts posted per night $b \in \{1, \ldots, 9\}$ (at most 12, the pool size); memory $\rho \in \{1, 0.75, 0.5, 0.25\}$; vote gate on and off; proof stop on; 1,000 episodes per setting (1,740 settings). Unless stated otherwise: $q = 6$, $q_c = 12$, $b = 1$ (the main setting, which posts about as many facts as Simulation 2 at rate 1), gate on. The vote gate is on throughout.

Summary. The 24 agents together own a proof of A0, and with full memory they keep it: no controller changes the swarm for more than a few days. **Forgetting is what makes the swarm steerable**, and in this simulation the effect peaks at intermediate memory: at $\rho = 0.5$ the false controller wins 78% of the votes (Figure 1). Seen side by side (Figures 2, 4–6), **susceptibility and readability move together**, while **the efficiency η moves much less** (0.56–0.91) and falls only where one controller grows much more than the other. **More budget pays only when agents remember well** ($\rho = 0.75$), and there it pays more the longer control lasts; elsewhere one fact per night is almost enough and the most effective per fact (Figures 3, 7, 8). **Reading more posts protects the proof when agents remember and helps the controller when they forget** (Figure 4). Every controller replaces part of the agents' own proofs (Figure 9).

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose vote on day t is answer a and $N_t = 24$ (every agent votes every day; before the first day nobody has voted and we set $x_{a,0} = 0$),

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

Paired susceptibility. Following `BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES` (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\mathrm{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G^{(0)}_t(\pi^+)$ for the controller targeting A0 and $G_t \equiv G^{(2)}_t(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\mathrm{A0}, \mathrm{A2}\}$ is drawn by a fair coin. The state X on day t is the vote of a randomly chosen agent, $X \in \{\mathrm{A0}, \mathrm{A1}, \mathrm{A2}\}$. Each episode e of arm s (silent 0, target A0 +, target A2 −) contributes its vote shares $y_s^{(e)}(x) = x_{x,t}$, and

$$P_s(x) = \frac{1}{E}\sum_{e=1}^{E} y_s^{(e)}(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I(Z;X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z;X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z;X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004; without it, by at most 0.005 where C is finite. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted on the nights before day t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring budgets,

$$\frac{\Delta G}{\Delta b} = \frac{G(b^{(k+1)}) - G(b^{(k)})}{b^{(k+1)} - b^{(k)}}. \tag{8}$$

Episode e uses the same random draws at every budget, so the slope is estimated from per-episode differences, $[x^{(e)}_{\mathrm{target},t}(b^{(k+1)}) - x^{(e)}_{\mathrm{target},t}(b^{(k)})]/(b^{(k+1)} - b^{(k)})$, whose mean is Eq. (8) and whose standard error gives the 95% intervals (typically ±0.006–0.02, about 30% narrower than for two independent means).

2 Votes over time

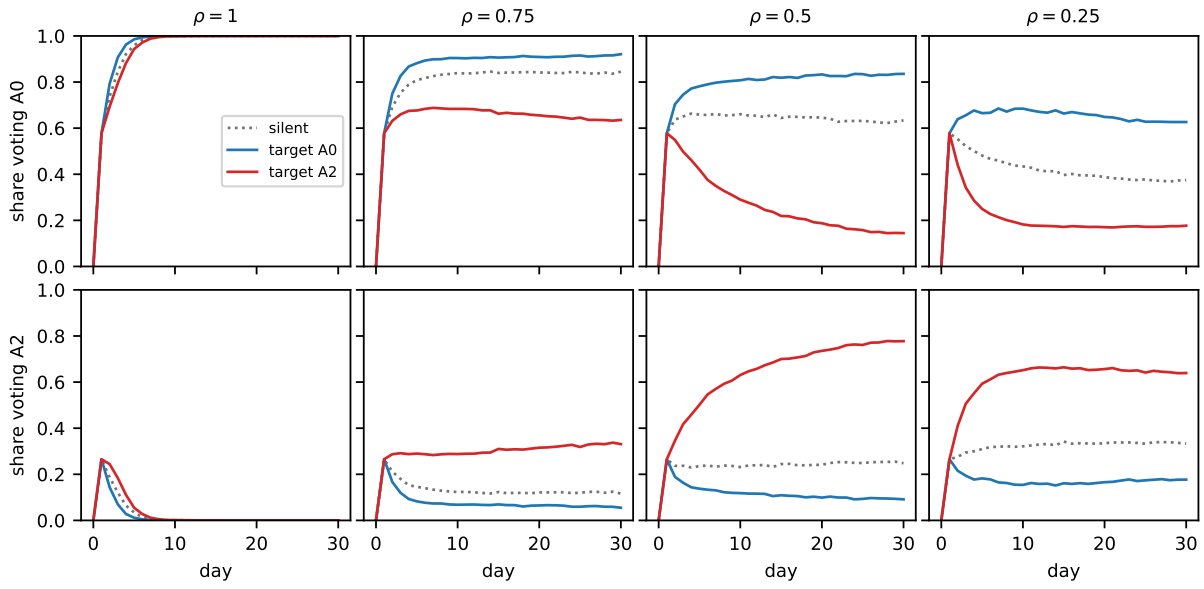


Figure 1: Share of agents voting A0 (top) and A2 (bottom) at the end of each day, Eq. (1), without a controller (dotted), with the controller targeting A0 (blue) and A2 (red); one column per memory ρ. $q = 6$, $q_c = 12$, $b = 1$, gate on. Day 0: nobody has voted yet, every share is 0. 95% intervals are narrower than the lines (±0.01).

- **The shorter the memory, the less firmly the silent swarm holds the truth**: 100%, 85%, 63% and 38% vote A0 on day 30 for $\rho = 1, 0.75, 0.5, 0.25$.
- **With full memory no controller matters**: every agent votes A0 by day ≈ 8 in all three arms.
- **The false controller wins the majority when** $\rho \leq 0.5$: A2 gets 78% ($\rho = 0.5$) and 64% ($\rho = 0.25$) of the votes, against 25% and 33% in silence; at $\rho = 0.75$ only 33%.
- **The truth controller restores the truth** to 92% ($\rho = 0.75$) and 84% ($\rho = 0.5$), but only to 63% at $\rho = 0.25$.

3 Observables as time series

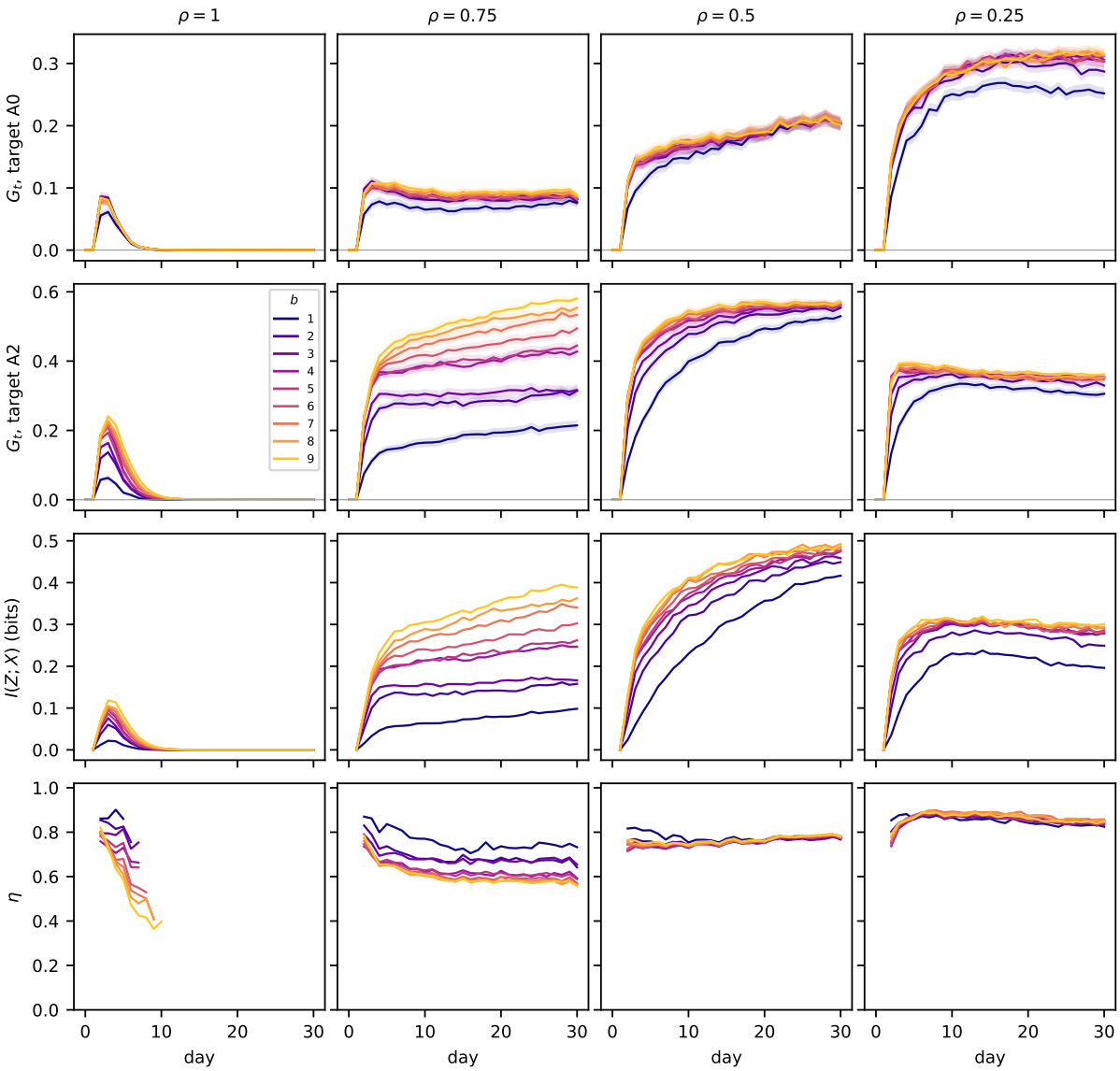


Figure 2: All observables over the days, one column per memory ρ, one curve per budget b; $q = 6$, $q_c = 12$, gate on. Rows: paired susceptibility G_t, Eq. (2), for target A0 and target A2 (bands: 95% intervals; $G_0 = 0$); readability $I(Z;X)$, Eq. (4); efficiency η, Eq. (6) (blank: undefined).

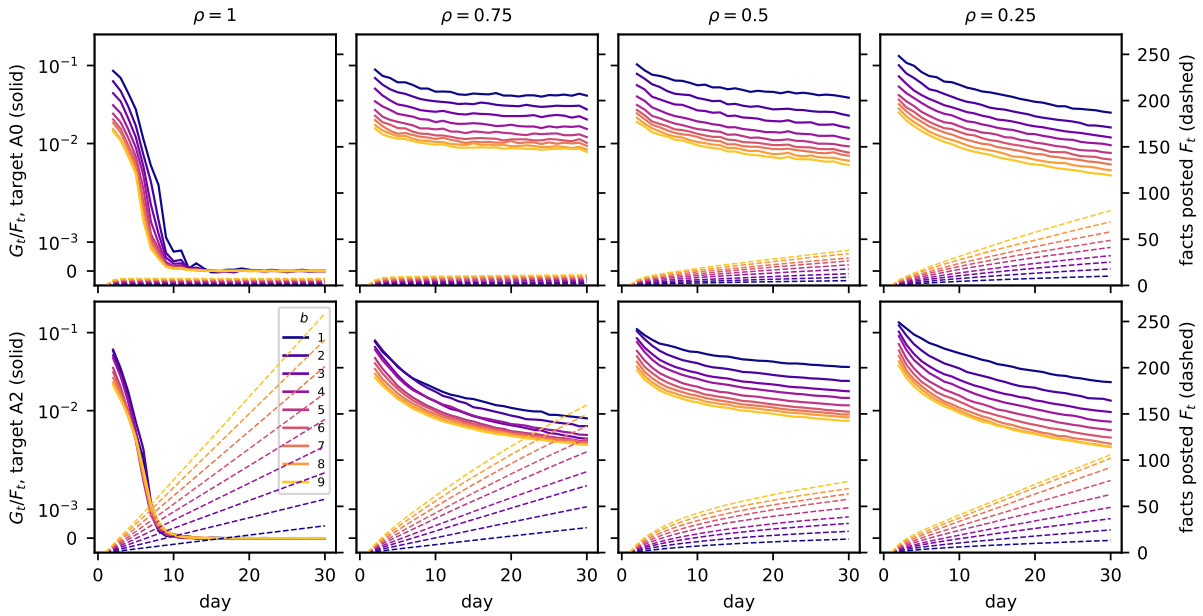


Figure 3: Gain per fact G_t/F_t (solid, left axis, log scale; shown once at least half a fact has been posted) and facts posted F_t (dashed, right axis), target A0 (top) and A2 (bottom), one colour per budget; $q = 6$, $q_c = 12$, gate on.

Susceptibility (Figure 2, rows 1–2).

- **It peaks at intermediate memory**: on day 30, target A2, $G = 0.22$–0.58 ($\rho = 0.75$), 0.53–0.57 ($\rho = 0.5$), 0.31–0.36 ($\rho = 0.25$); target A0, 0.08–0.09, 0.20, 0.25–0.32.
- **With full memory it is only transient**: target A2 reaches 0.24 on day 3 ($b = 9$) and is gone by day ≈ 12.
- **The budget matters only when agents remember well**: at $\rho = 0.75$, target A2 grows from 0.22 ($b = 1$) to 0.58 ($b = 9$); at $\rho \leq 0.5$ every $b \geq 2$ gives about the same.
- **With heavy forgetting a large budget overshoots**: at $\rho = 0.25$, $b = 9$, target A2 peaks at 0.39 (day 5) and settles at 0.36.

Readability and efficiency (rows 3–4).

- **Readability has the shape of the susceptibility**: it is largest at $\rho = 0.5$ ($I = 0.42$–0.49 bits on day 30), grows with the budget at $\rho = 0.75$ ($0.10 \to 0.39$) and is transient with full memory.
- **Efficiency moves much less**: after day 3, $\eta = 0.82$–0.90 ($\rho = 0.25$), 0.73–0.82 ($\rho = 0.5$), 0.56–0.86 ($\rho = 0.75$). It falls with the budget only at $\rho = 0.75$, where the false controller grows with b while the truth controller is near its ceiling: η reports how symmetric the two controllers' effects are, not their size.

Gain per fact and facts posted (Figure 3).

- **Each fact buys most in the first days**: G_t/F_t is largest early and falls as facts accumulate.
- **The smallest budget is the most effective per fact**: on day 30, target A2, $\rho = 0.5$: 0.036 per fact at $b = 1$ (15 facts) against 0.007 at $b = 9$ (77 facts); at $\rho = 0.25$: 0.023 against 0.003.
- **The facts posted grow almost linearly with b for the false controller** (up to 160 by day 30 at $\rho = 0.75$); the truth controller posts only 2–81, because the gate keeps it silent once the swarm agrees, which is why it gets more per fact (0.039 at $\rho = 0.5$, $b = 1$).

4 Phase diagrams

All panels on day 30, gate on; the susceptibility is for target A2 (grey number: its 95% interval includes 0); readability and efficiency are for the pair of targets. Colour scales are shared with the Simulation 2 report.

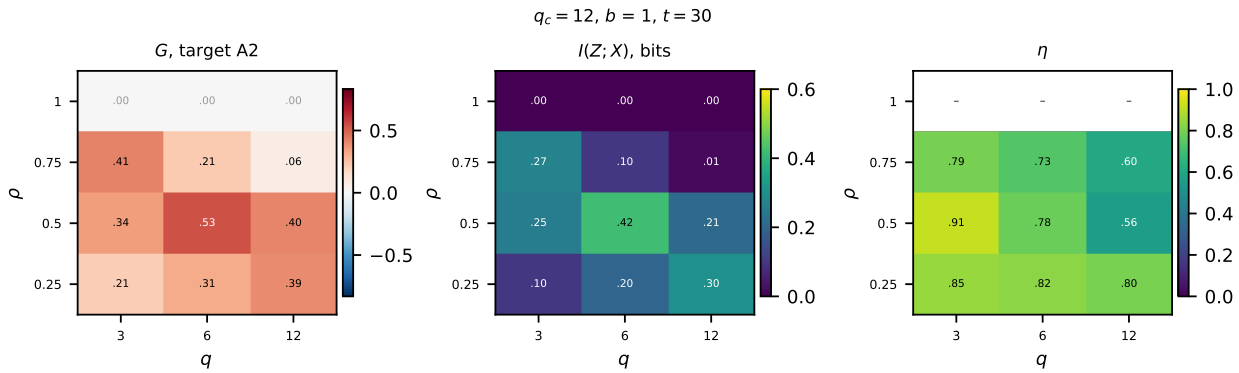


Figure 4: Susceptibility, readability and efficiency over memory ρ and the posts each agent reads q; $q_c = 12$, $b = 1$.

- **Reading more defends the proof only when agents remember well**: at $\rho = 0.75$, G falls from 0.41 ($q = 3$) to 0.06 ($q = 12$) and I from 0.27 to 0.01 bits; at $\rho = 0.25$, G rises from 0.21 to 0.40 and I from 0.10 to 0.30. A likely reason, not tested: with heavy forgetting the proof cannot be rebuilt anyway, and reading more only spreads the controller's facts faster.
- η **stays at 0.56–0.91**, unrelated to this reversal.

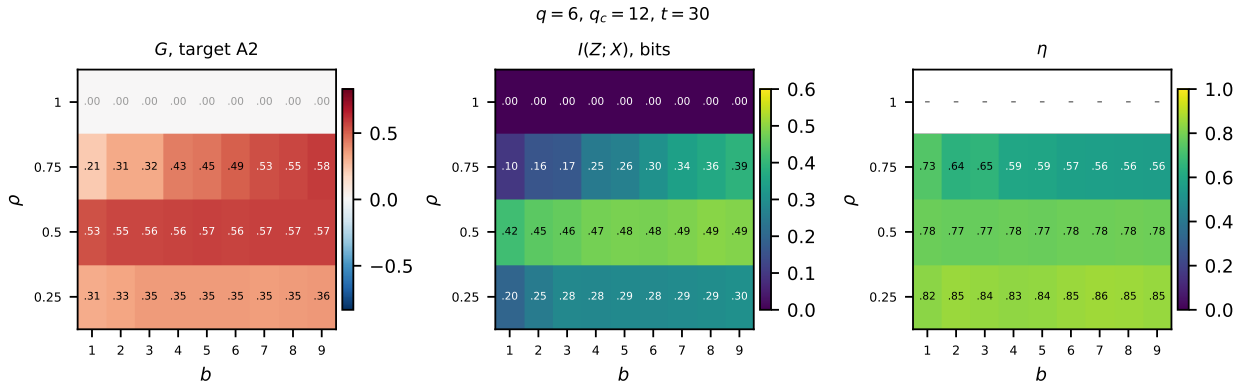


Figure 5: Susceptibility, readability and efficiency over memory ρ and the budget b; $q = 6$, $q_c = 12$.

- **Memory sets the level**: $G = 0$ at $\rho = 1$, largest at $\rho = 0.5$ (0.53–0.57); the budget matters only at $\rho = 0.75$ (0.22 to 0.58).
- G **and** I **move together** (I from 0.10 to 0.49 bits); η **moves the other way at** $\rho = 0.75$ (0.73 to 0.56 as b grows), where the pair turns lopsided.

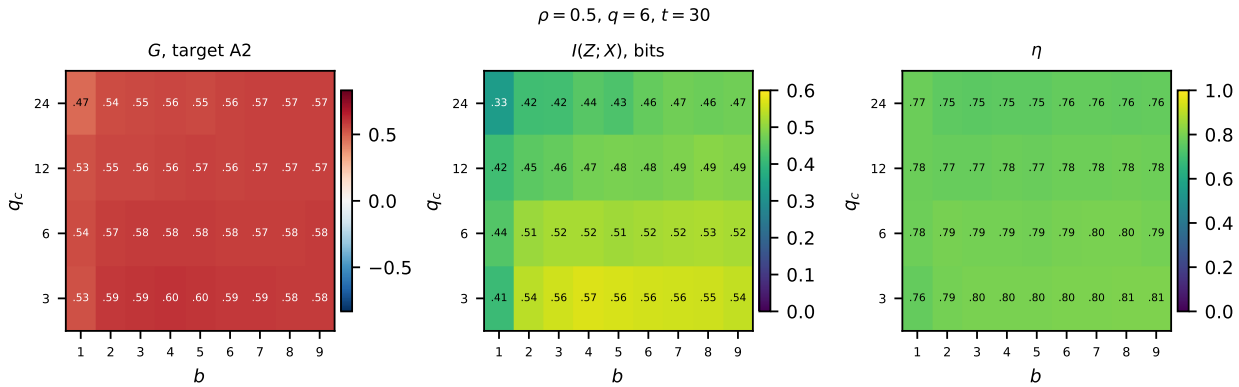


Figure 6: Susceptibility, readability and efficiency over the posts the controller reads q_c and the budget b; $\rho = 0.5$, $q = 6$.

- **At $\rho = 0.5$ neither the controller's reading nor its budget matters much**: $G = 0.47$–0.60 (mean 0.58, 0.57, 0.56, 0.55 for $q_c = 3, 6, 12, 24$); $I = 0.33$–0.57 bits; $\eta = 0.75$–0.81.
- **Reading fewer posts helps slightly**, likely because with few posts read the vote gate silences the controller less often (not checked).

5 Susceptibility against the budget, and its slope

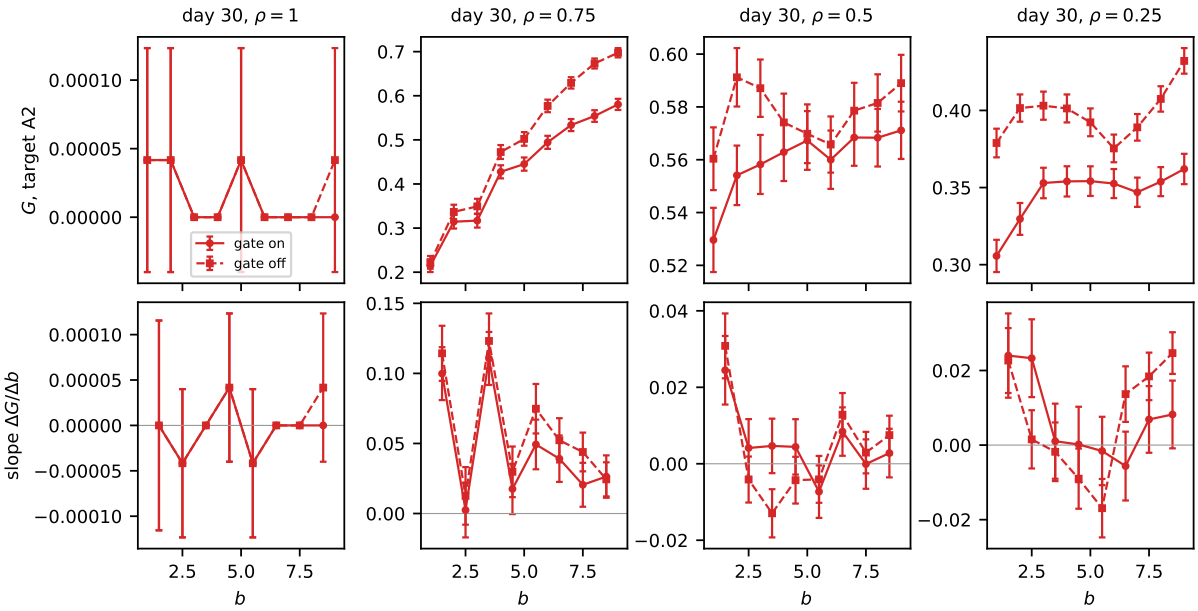


Figure 7: Top: G on day 30, target A2, against the budget b, one panel per ρ; gate on (solid) and off (dashed); bars: 95% intervals. Bottom: the slope between neighbouring budgets, Eq. (8), with paired 95% intervals.

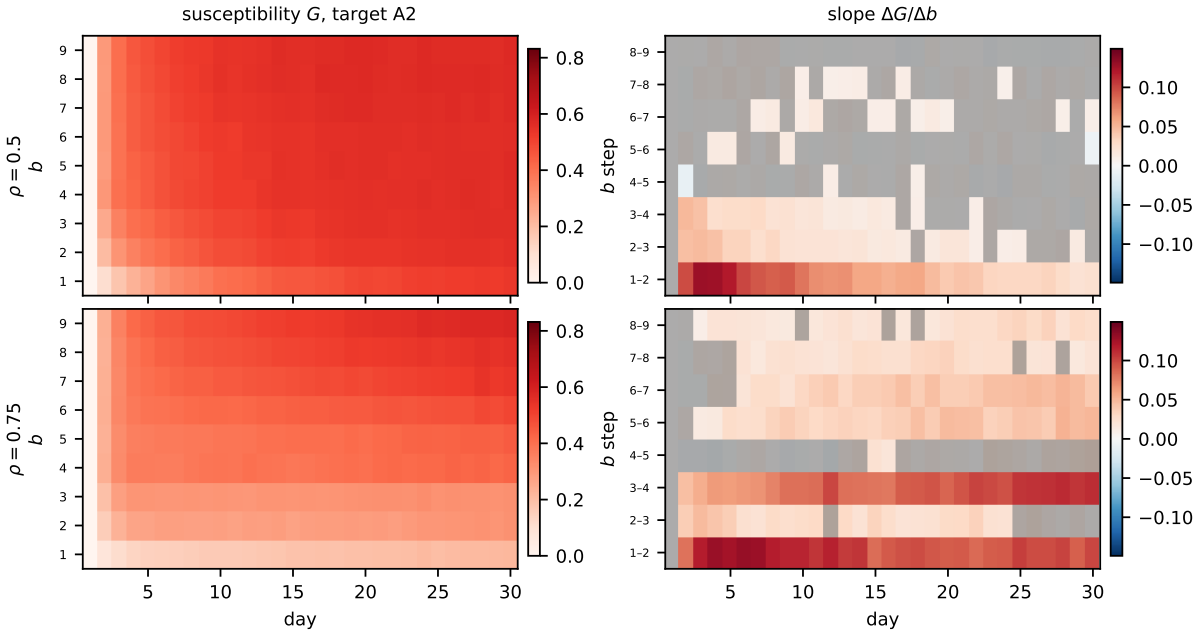


Figure 8: G, target A2 (left), and its slope between neighbouring budgets (right) over the days (x) and the budget (y), for $\rho = 0.5$ (top) and 0.75 (bottom); $q = 6$, $q_c = 12$, gate on. Each row of the right panels is one step between neighbouring budgets; faded cells: its paired 95% interval includes 0.

- **When agents remember well, more budget keeps buying susceptibility** (Figure 7): at $\rho = 0.75$ the slope is 0.10 for the first step and still 0.04 on average after it; at $\rho = 0.5$ and 0.25 it is 0.02 for the

first step and about 0.002 after it.

- **At $\rho = 0.5$ the slope fades with time** (Figure 8): the first step falls from 0.12 (day 5) to 0.02 (day 30) as $b = 1$ catches up; $b = 9$ reaches 90% of its day-30 value by day 7, $b = 1$ by day 19.
- **At $\rho = 0.75$ it does not**: on day 30 most steps are still significantly positive ($b = 1 \to 2$: 0.10; $3 \to 4$: 0.11; $5 \to 9$: 0.02–0.05), and the step from $b = 3$ to 4 grows from 0.05 (day 2) to 0.11 (day 30). More budget pays more the longer control lasts.
- **The returns come in uneven steps**: at $\rho = 0.75$ the steps $2 \to 3$ and $4 \to 5$ add nothing measurable while their neighbours add 0.05–0.11. A likely reason, not tested: the best set of b facts from the 12-fact pool changes composition only at some values of b.
- **Turning the gate off adds susceptibility where the budget matters**: up to +0.12 at $\rho = 0.75$.

6 Who holds the proof

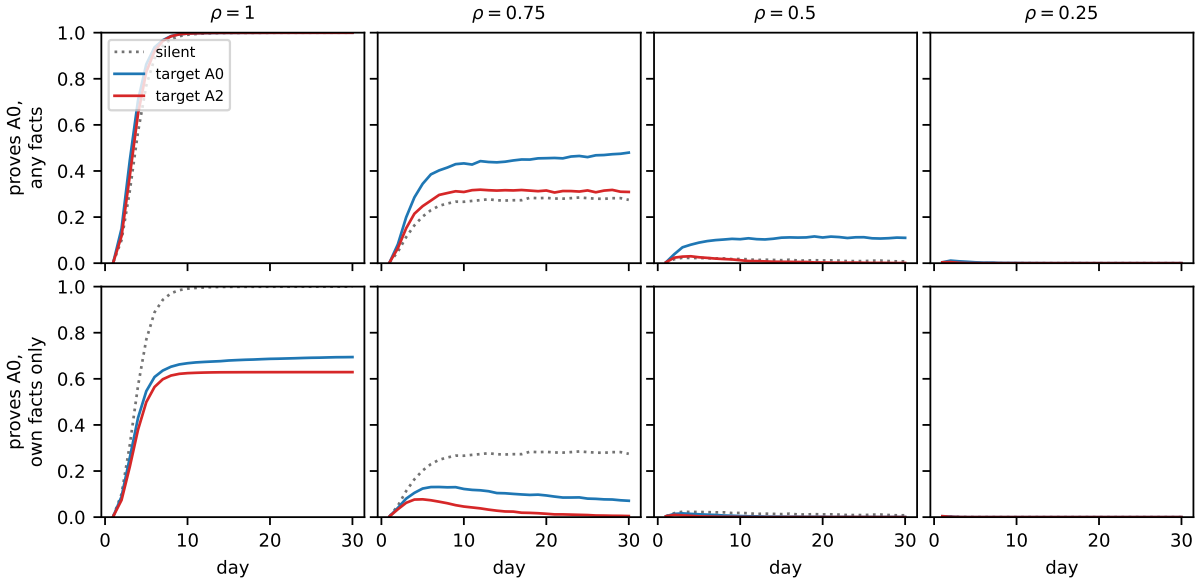


Figure 9: Share of agents whose remembered facts prove A0 (top) and prove it from the agents' original facts only (bottom); silent (dotted), target A0 (blue), target A2 (red); $q = 6$, $q_c = 12$, $b = 1$, gate on.

- **Proofs need memory**: on day 30 the silent swarm has 100%, 28%, 1% and 0% of agents proving A0 for $\rho = 1, 0.75, 0.5, 0.25$. Below $\rho = 0.75$ votes rest on beliefs, not proofs.
- **Every controller replaces the agents' own proofs**: with full memory all agents prove A0 under any arm, but only 69% (target A0) and 63% (target A2) from their own facts (100% in silence); at $\rho = 0.75$, 48% prove A0 under the truth controller but only 7% from their own facts.
- **The false controller's true facts complete proofs of A0** (31% against 28% in silence at $\rho = 0.75$), while own-evidence proofs fall to 0.5%.

7 Conclusions

1. **Memory decides whether a swarm that owns a proof can be steered.** With full memory every effect vanishes within about ten days. As memory shortens the silent swarm holds the truth less firmly (100%, 85%, 63%, 38% voting A0 on day 30), and the susceptibility peaks at $\rho = 0.5$.
2. **Susceptibility and readability move together; efficiency does not.** In every time series and phase diagram G and I rise, fall and reverse together; η stays at 0.56–0.91 and falls only where one controller grows much more than the other.
3. **More budget pays only when agents remember well**, and there it pays more the longer control lasts; elsewhere one fact per night is almost enough and the most effective per fact.

4. **The posts each agent reads switch sides with memory**: they defend the proof at $\rho = 0.75$ and help the controller at $\rho = 0.25$, in susceptibility and readability alike.
5. **Control replaces the agents' own proofs**, even when every agent ends up proving the truth.
6. **Compared with Simulation 2** (same settings, about the same facts posted): Simulation 1 is more susceptible at $\rho = 0.5$ (0.53 against 0.33, target A2) and less at $\rho = 0.25$ (0.31 against 0.43).

Caveats. One world, 24 exact-Bayesian agents voting by probability matching, one controller policy posting a fixed batch each night; 1,000 episodes per setting (95% intervals on G about ± 0.02). Information is plug-in from 1,000 episodes per arm. Data: `results/simulation_1/2026-10-15_sim1_pm_symmetric_fine`; tables and figures: `analysis/task003_llm_free/shared_analysis/gold_standard/`.